

Low-Cost Open-Source Absorbance Spectrophotometry on a Monochrome Global-Shutter Camera

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Abstract—We present MonoSpectro, an open-source visible-range absorbance spectrophotometer built from a monochrome global-shutter camera and a 1000^{-1}mm diffraction grating bonded directly to the lens, for about €166 in components, together with a calibration transfer that makes its absorbance readings quantitatively comparable with those of a commercial benchtop instrument. The correction expresses the deviation between the two instruments as a wavelength-dependent gain and offset, expands both to first degree, and fits them in a single least-squares pass over every wavelength and every calibration sample — four coefficients for the whole spectrum. Fitted on nine food colourings and evaluated once on four laboratory dyes measured in a later session, it reduces the root-mean-square error against the reference by 72 % over 420 nm to 780 nm, and improves every sample. Fitting the same relation independently at every wavelength, with 180 times as many coefficients, recovers only 54 %: with few calibration samples, constraining the model is what makes it generalise. All hardware designs, firmware and analysis code are released under the MIT licence.

Index Terms—Calibration transfer, absorbance spectroscopy, CMOS image sensor, low-cost instrumentation, open-source hardware, water quality monitoring.

I. INTRODUCTION

Absorbance spectrophotometry is a staple measurement across chemistry, biology and environmental monitoring, but laboratory-grade accuracy arrives bundled with cost, bulk and mains power. That rules it out precisely where sustained measurement is most valuable: unattended and in the field, on a buoy or a moored station [1]–[3]. A substantial literature of low-cost and open-source photometers has grown in response [4]–[6], and consumer image sensors in particular have become credible detectors for quantitative work, in purpose-built spectrometers [7]–[9] and in smartphone photometry validated on real analytes [10]–[13]. MonoSpectro builds directly on that line of work.

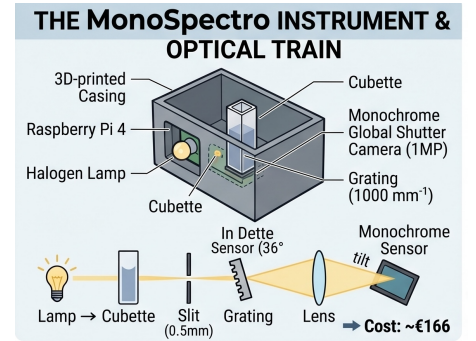
What is less often demonstrated is that the numbers such an instrument produces are quantitatively comparable with those of a commercial instrument *on samples its calibration never saw*. Many published validations fit and evaluate on the same set of solutions [7], [9], [12], which measures how well a model reproduces its own training data rather than whether it holds on new chemistry. Where readings are to be pooled

across instruments and sites, that is the property that matters [14], [15].

In analytical chemistry this is the calibration-transfer problem, studied for decades in near-infrared spectroscopy [16]–[18]: direct and piecewise direct standardisation map spectra recorded on one instrument onto those of another [19], [20], and slope-and-bias schemes do the same with far fewer degrees of freedom [21], all of them normally fitted on tens of transfer standards. This letter asks what transfers when there are nine. We model the deviation as a wavelength-dependent gain and offset, constrain both to a first-degree expansion, and test the result once on dyes measured in a later session, comparing not against the uncorrected instrument but against a freer version of the same model.

II. THE PLATFORM

The optical train is deliberately minimal: a 10 W halogen lamp, a standard 10 mm cuvette, a 0.5 mm entrance slit, and the wide-angle M12 lens supplied with the camera, carrying a 1000^{-1}mm transmission diffraction grating film bonded to its front surface. A collimator is not indispensable in this geometry: the slit sits 60 mm from the lens (Fig. 1a), far



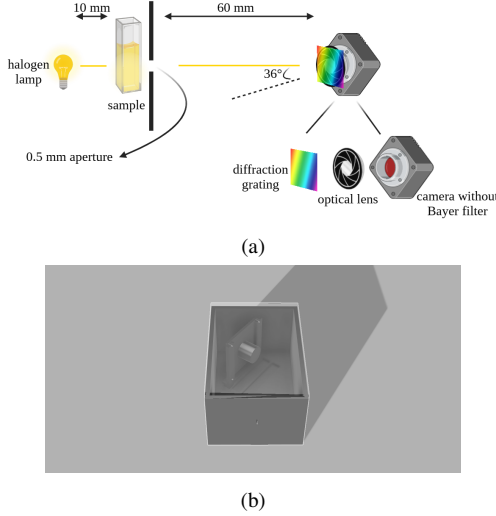


Fig. 1. (a) Optical path. Light from the halogen source crosses the cuvette and the entrance slit and is dispersed by a diffraction grating bonded directly to the front face of the lens, onto a sensor tilted by 36° . (b) The assembled instrument. [replace with the hero photograph of the built unit]

enough that the beam reaching the grating is close to parallel. For a transmission grating of period d the first-order maxima satisfy

$$m\lambda = d(\sin \theta_i + \sin \theta_m). \quad (1)$$

Equation (1) fixes the two geometric choices visible in Fig. 1a. The first is the tilt. The manufacturer specifies a dispersion angle of 36° for this film, and Eq. (1) identifies what that angle is: at normal incidence $\sin \theta_1 = \lambda/d$, so 36° is the first-order direction for 589 nm, near the middle of the visible band. The camera assembly is tilted by exactly that angle from the slit axis, which centres the band on the sensor. The second is the lens. Over 400 nm to 800 nm the same relation swings the first order from 23.6° to 53.1° , so the lens has to accept 29.5° of field around the tilt direction; the stock wide-angle lens, specified at 110° horizontally, covers that with room to spare, and its short focal length is what keeps the whole first order inside a 1/4-inch sensor.

The detector is a 1 Mpixel monochrome global-shutter sensor read out at 1280×720 . Monochrome matters: with no Bayer mosaic every pixel is an unfiltered intensity reading, and no colour interpolation stands between the light and the numbers [22]. It is read at 8 bits, so exposure must bring the brightest part of the blank close to saturation without reaching it, and it carries no infrared-cut filter.

Acquisition runs on a single-board computer (SBC), a Raspberry Pi 4. Frames are captured through the picamera2 library; every column of a user-defined region of interest is averaged vertically to give one intensity value per pixel column; a low-order polynomial maps pixel position to wavelength, following the procedure described in [23]; and absorbance is reported against a stored blank as

$$A = \log_{10}(I_0/I), \quad (2)$$

TABLE I
SUMMARY OF COMPONENTS.

Item	Specification
Camera	Ardurcam OV9281, 1 MP monochrome
Lens	M12 wide angle, 130° diagonal
Grating	Edmund #4621, 1000^{-1} mm film
Light source	Osram Halostar 64415 S, 10 W
Cuvette	Plastic, 10 mm path length
Compute	Raspberry Pi 4
Enclosure	3D printed, opaque filament
Total cost	$\approx \text{€}166^*$

*Components only; the reference instrument is not included.

where I_0 is the blank intensity and I that of the sample at the same wavelength. The interface is served over HTTP, and the Pi can raise its own wireless access point, so the instrument needs no infrastructure in the field. Table I lists the components and Fig. 1b shows the assembled unit.

All reference measurements were made on a K LAB Optizen Alpha double-beam UV-Vis spectrophotometer: Czerny–Turner monochromator with a 1200^{-1} mm blazed grating, tungsten-halogen and deuterium sources, silicon photodiode detection, 1 nm spectral bandwidth over 190 nm to 1100 nm, wavelength accuracy ± 0.3 nm, and photometric accuracy ± 0.002 at 0.5 absorbance units. It plays the part of the master instrument in the sense of the standardisation literature [18], [19]: the quantity MonoSpectro is asked to reproduce is what this instrument reads.

III. CALIBRATION TRANSFER

The instrument's absorbance readings deviate from the reference in a way that varies with wavelength but is stable across samples. That deviation arises from the spectral response of the detector, the transmission of the optics and the stray-light floor of the enclosure [22], [24], none of which depend on what is in the cuvette, so it is a property of the *instrument* rather than of the samples used to measure it. We therefore model it directly, as a linear relation at each wavelength,

$$A_{\text{ref}}(\lambda) = a(\lambda) A_{\text{dev}}(\lambda) + b(\lambda). \quad (3)$$

What matters is not Eq. (3) itself, which is fixed, but how much freedom $a(\lambda)$ and $b(\lambda)$ are given when they are estimated. Estimating them independently at each wavelength and smoothing the resulting curves afterwards is the established approach, and the one direct standardisation takes [19], [20]. Over the working range, sampled at 1 nm, that is two coefficients at each of 361 wavelengths, 722 in all, estimated from nine samples.

The alternative adopted here constrains both curves to first-degree polynomials in wavelength — in a Chebyshev basis, which stays numerically well conditioned over a fixed interval [25] — and fits them in a single least-squares pass that uses every wavelength and every training sample at once, so smoothness belongs to the model instead of being a filter applied afterwards. A first-degree fit makes $a(\lambda)$ and $b(\lambda)$ straight lines and the whole correction four numbers. The

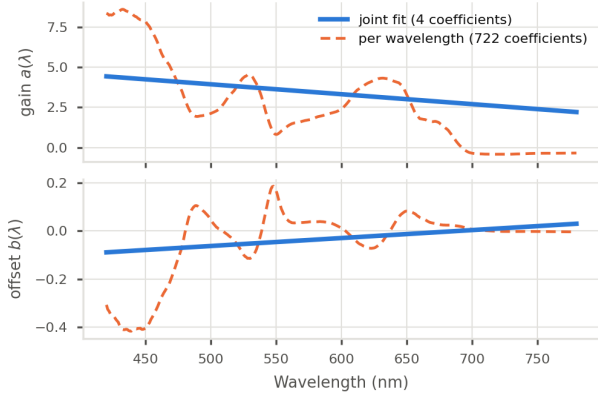


Fig. 2. The fitted correction. The per-wavelength fit is free to follow the training samples, while the first-degree joint fit is two straight lines. The constrained model keeps the drift and drops the excursions.

degree is set rather than searched because the optics give no reason for either curve to be complicated: what they have to capture is a gentle drift with wavelength. Searching confirms it from the other side — higher degrees scored marginally better in cross-validation, which rewards flexibility, and then did worse on the test set. The wavelength range is still chosen by leave-one-out on the training set alone. The same form of correction is used for the camera platform in [23], where it is fitted and reported over a different sample set. Fig. 2 shows what the two versions look like once fitted.

Three further steps belong to the data processing rather than to the model. Both sides are passed through a Savitzky–Golay filter [26] before the correction is fitted or applied, because the correction is pointwise and noise is amplified wherever $a(\lambda) > 1$; a window swept between 25 and 85 nm puts the error minimum between 45 and 55 nm, and 45 nm — the least smoothing that reaches the plateau — is used. Since Eq. (2) has a hard floor at zero, the corrected output is constrained to be non-negative, without which the fitted line drifts below zero in the near-baseline regions where both the gain and the training data are small and noisy. That constraint is imposed as a rounded floor rather than a clip: over a transition of fixed half-width ϵ — a processing constant, not one of the four fitted coefficients — the output is blended into zero by a quadratic matching both value and slope at each join. Scored on the near-zero region where it acts, error falls from 0.024 at a hard clip to 0.021 at $\epsilon = 0.15$.

IV. EXPERIMENTAL PROTOCOL

The training set comprised nine aqueous solutions prepared from a commercial twelve-colour concentrated liquid food-colouring set (6 mL per bottle; Amazon ASIN B08X6FMHKP), each made by diluting one drop of dye in 10 mL of distilled water, which spreads absorbance across the visible band, much as organic dyes are used to characterise low-cost photometers elsewhere [12]. The manufacturer discloses neither the identity nor the concentration of the colorants, so they are named here by their commercial colour names; their reference spectra are included with the analysis code, so that they can be recognised independently of the

TABLE II
TEST-SET PERFORMANCE. RMSE IS IN ABSORBANCE UNITS, PER SAMPLE AND AVERAGED. R^2 IS POOLED OVER ALL FOUR SAMPLES AND ALL 361 WAVELENGTHS, THAT IS 1444 POINTS, AND N_c IS THE NUMBER OF FITTED COEFFICIENTS.

	Uncorrected	Per wavelength	Joint, 1st degree
Bromothymol blue	0.440	0.099	0.149
Congo red	0.440	0.180	0.099
Rhodamine B	0.389	0.233	0.064
Methylene blue	0.116	0.129	0.073
Mean RMSE	0.346	0.160	0.096
Pooled R^2	0.149	0.825	0.936
N_c	0	722	4

supplier. Across the nine, the device tracks the position and width of every band but reads systematically low on the absorbance scale, which is the deviation the response function models.

The test set comprised four laboratory-grade dyes, each prepared in a 100 mL volumetric flask with distilled water as the solvent: methylene blue (1.1 mg), rhodamine B (1.4 mg), bromothymol blue (4.5 mg, measured in its acid form) and congo red (1.6 mg), giving 11, 14, 45 and 16 mg^{-1}L respectively. These solutions were prepared and measured in a session later than the training set; both sessions used the same optical configuration and the same procedure, so the test set is separated from the training set by session and by chemistry alone. Every choice in the correction — degree, wavelength range, smoothing window — was fixed on the training set before these four dyes were processed, and they were scored once against the reference; no result below was used to revise the model.

Each reported spectrum is the mean of ten consecutive readings of a single preparation, taken in plastic cuvettes of 10 mm path length, at a stable ambient temperature of 25 °C and after a five-minute lamp warm-up on both instruments. The reference instrument is double-beam and so carries its blank continuously; on MonoSpectro the same blank of distilled water, in the same cuvette, was re-taken between samples. Agreement is quantified as the root-mean-square error (RMSE) between the device and reference spectra over 420 nm to 780 nm, and is reported throughout beside the do-nothing baseline of applying no correction at all.

V. RESULTS AND DISCUSSION

Table II and Fig. 3 give the test-set results. The joint fit lowers the mean RMSE from 0.346 to 0.096 absorbance units (72 %) and improves every dye, by between 37 % and 84 %; of that gain, input smoothing alone accounts for about two thirds, and the non-negativity constraint and rounded floor for the rest. The per-wavelength fit, with 722 coefficients instead of four, stops at 0.160 and a pooled R^2 of 0.825 against 0.936. Its extra freedom lets it follow the particular features of nine training samples: in earlier versions this pulled methylene blue, which already agreed well with the reference, toward the average of the training set, and it is also why cross-validation, which rewards flexibility, was not used to choose the degree.

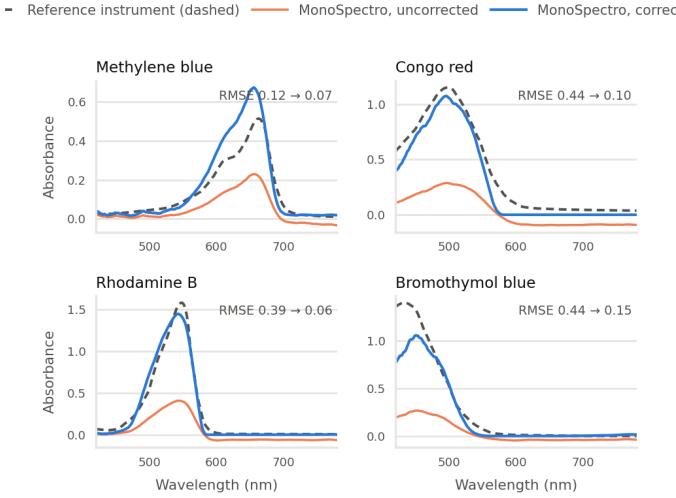


Fig. 3. The test set: four laboratory dyes measured in a later session, shown uncorrected, corrected, and as recorded by the reference instrument (dashed).

The spectral shape is recorded well even before correction: a single scale factor per dye gives R^2 between 0.89 and 0.98, and every band maximum lies within 18 nm of the reference. What the response function corrects is the absorbance scale, not the band positions.

The corrected spectra are nonetheless not laboratory-grade: a mean error of 0.096 is some fifty times the photometric accuracy of the reference. The weakest region is below 500 nm, where a 2800 K halogen source emits little light and the silicon response is falling; bromothymol blue, which absorbs there, is the only dye the joint fit improves less than the per-wavelength fit, while methylene blue is overshot at its peak. Three uncharacterised effects contribute. The wavelength axis is approximate: Eq. (1) is non-linear in pixel position, the wide-angle lens distorts away from its axis, and the tilted sensor varies the magnification, all of which a rational pixel-to-wavelength map would describe better than the present polynomial. Stray light reaches the sensor through the open enclosure, which has no baffles or order-sorting filter, compressing absorbance at high optical density [24], [27], and the finite bandwidth does the same on narrow bands [28], beyond 700 nm outside the grating's specified range. Precision across preparations is unquantified, since each spectrum averages ten readings of a single preparation. The most direct improvements follow from this: a source with more blue output, a training set that absorbs strongly below 500 nm, and more training samples, so that the response function is constrained where it now extrapolates. For a portable instrument two orders of magnitude cheaper than the reference, the results support a narrow claim — that the absorbance scale transfers well enough to make readings comparable — rather than a metrological characterisation.

VI. CONCLUSION

Four coefficients describing a wavelength-dependent gain and offset are enough to transfer the absorbance scale of a €166 camera spectrophotometer onto that of a commercial benchtop instrument, on samples outside the calibration set.

With nine training pairs, constraining the model is what makes it generalise, the same lesson the standardisation literature draws from much larger transfer sets [16], [18]. Improving the wavelength mapping and the blue end of the spectrum would let a single correction be fitted once and reused across units, which is the precondition for pooling field measurements across instruments and sites.

ACKNOWLEDGMENT

DATA AND CODE AVAILABILITY

The enclosure designs, the instrument firmware, the calibration and analysis tools and the scripts that generate every figure in this letter are available as supplementary material at <https://github.com/alexmanuel27/MonoSpectro>, under the MIT licence.

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